

Introduction

I am an enthusiast of problem solving. Always seeks for best solution of problems encountered in life, work, or researches in spare time. Good at using debug tools and working with large codebase with mixture of several frameworks or tools to solve difficult problem.

Education

Bachelor of Computer Science and Information Engineering

2020/09 – 2025/06

National Taiwan Normal University

Work Experience

Software Engineer

2025/08 – Now

Graid Technology

#CUDA #Linux Kernel

- Debug Linux kernel to fix synchronization problem in kernel module under QEMU environment.
- Develop RAID system based on GPU hardware.
 - Develop CUDA kernel to accelerate RAID.
 - Develop Linux kernel driver for RAID software.

Embedded Software Engineering Intern

2025/05 – 2025/07

Apple

#Objective-C #Python

- Develop internal flash tool in Objective C without prior knowledge.
- Develop internal web tool, and migrate local secret to cloud infrastructure.

Software Engineer Intern

2023/12 – 2025/05

Appier

#Python #Kubernetes #Jenkins #Trino #GCP #Databricks

- Optimize Trino database by using bit vectorized query.
 - Refactor and optimize the SQL statement generator. Leads to the following impact:
 - Saves 56% of CPU time.
 - Saves 39% of wall time.
 - Understanding existing Python codebase that cooperate with of Spark (Scala), Airflow, Trino.
 - Migrate complex old PR to newer version of codebase.

Side Projects & Open Source Contributions

ggml: <https://github.com/jw910731/ggml/tree/tt-support>

2026/06

Tensor library for machine learning

#C++ #C #AI Inference

- Support running Z-Image Turbo on Tenstorrent Blackhole using stable-diffusion.cpp.
- Analyze performance bottleneck to ensure workload can be done in a reasonable time.
- Test driven development to ensure correctness of implemented operators.

envbuilder: <https://github.com/coder/envbuilder>

2024/03

Devcontainer support for Cloud IDE in Kubernetes

#Golang #Kubernetes

- Add registry cache mirror control environment variable.
- Avoids DockerHub rate limits.

HIME: <https://github.com/hime-ime/hime>

2021/03

Open source input method engine

#C #Dynamic Linking #GDB #QT Debug

- Discover a case where the IME cannot work.
- Use gdb and QT debug environment variable to find root cause.
- Fix by linking missing library.

rtl8852au-openwrt: <https://github.com/jw910731/rtl8852au-openwrt>

2025/05

Port realtek WiFi adapter out-of-kernel driver to OpenWRT

#C #Backports #OpenWRT

- Patch the driver to fix compile error caused by Linux backports
- Integrate with build system of OpenWRT

Skills

Programming	C/C++, Java, Python, Golang, JavaScript, PHP, Rust
Infrastructure & DevOps	GitHub Action, Jenkins, Docker, Nginx, Kubernetes
Communication	Chinese (native), English
Other	Linux System , Linux Admin, MySQL, LaTeX, Typst